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Translate Pharmacokinetics of PD-1/PD-L1 Monoclonal Antibodies from Cynomolgus Monkey to Human: Comparison of Different Approaches



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ABSTRACT

Antibodies blocking programmed death-1 (PD-1) and its natural ligand programmed death-ligand 1 (PD-L1) have been proved to be promising strategies in recent years. Hundreds of PD-1/PD-L1 antibodies are under development worldwide. Prediction of human pharmacokinetics (PK) in the preclinical stage is critical for designing dosing regimens in first-in-human studies. This study aims to predict the PK of PD-1/PD-L1 antibodies in human by scaling of monkey data. A systematic literature search of published articles on the PK of PD-1/PD-L1 antibodies in cynomolgus monkey and in human was conducted. Allometric scaling (AS), the species time-invariant (STIV) method, as well as physiologically based pharmacokinetic (PBPK) modeling were investigated. Six antibodies (avelumab, atezolizumab, nivolumab, pembrolizumab, cemiplimab, and zimberelimab) were included for investigation. The exponents used in this study were 0.85 and 1 for clearance (CL) and distribution volume (V), respectively, both for AS and STIV methods. The generic PBPK model for macromolecules in PK-Sim was used without further modifications. The dissociation constant of the antibody for binding to FcRn (K_D) in endosome space for human was assumed to be two-fold of that for monkey. Predicted human CLs for the majority of drugs were within the observed range, while Vs were not well predicted using the AS method. The STIV method and the generic PBPK model can be employed to translate concentration-time curves of PD-1/PD-L1 antibodies from cynomolgus monkey to human with comparable efficacy. The results of this study provide reference for the early development of PD-1/PD-L1 antibodies.

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Introduction

Cancer immunotherapy has been proved to be a promising strategy in recent years. Among a variety of targets, programmed death-1 (PD-1) and its natural ligand programmed death-ligand 1 (PD-L1) is a dominant immune checkpoint pathway operative in the tumor microenvironment.¹ Accumulating evidence indicates that antibodies blocking PD-1 or PD-L1 have remarkable clinical efficacy in the treatment of various solid and hematologic malignancies.² Despite the approved drugs, hundreds of clinical trials of PD-1/PD-L1 antibodies are ongoing worldwide.^{3,4}

Prediction of human pharmacokinetics (PK) in the preclinical stage is critical for designing dosing regimens in first-in-human studies. Several species, such as mouse, rat and dog are usually used for scaling animal PK to human for small molecules. Different to small molecules, systemic clearances of monoclonal antibodies do not rely

on hepatic metabolism, and antibodies cannot be cleared in an intact form by renal glomerular filtration. Due to the similar neonatal Fc receptor (FcRn) binding affinity against antibodies, cynomolgus monkey is the most widely used species to predict the PK of monoclonal antibodies in human.⁵ Several empirical approaches for predicting human PK from monkeys have been proposed and used for therapeutic antibodies. Allometric scaling (AS), based on a power law relationship between body size and physiological and anatomical parameters, is widely used due to simplicity and historical confidence.⁶ The species time-invariant (STIV) method, proposed by Dedrick in 1973, is another similar approach which relies on the principles of allometry and physiological time in a species for the prediction of human PK.^{7,8} Furthermore, physiologically based pharmacokinetic (PBPK) models have also been more and more widely used in the process of drug development.⁹ A generic whole-body PBPK model for therapeutic proteins in PK-Sim was proposed by Niederaht et al., and has become part of the Open Systems Pharmacology Suite.¹⁰ In recent years, the generic PBPK model has been validated by several researchers and shown its potential in predicting human PK.^{11,12}

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Although several studies have summarized or compared the prediction of human PK using preclinical data based on a number of proteins,^{8,13,14} few PD-1/PD-L1 antibodies have been included in their database, and the generic PBPK model in PK-Sim has not been investigated. This study aims to evaluate the empirical AS method, the STIV approach, and the generic PBPK model in PK-Sim for scaling monkey PK data to human, thus to provide reference for the early clinical development of PD-1/PD-L1 antibodies.

Methods

Data

Cynomolgus monkey and human PK data were obtained from literature. Firstly, a systematic literature search of published articles on PD-1/PD-L1 antibodies in cynomolgus monkey was conducted in the online PubMed database. Relevant papers were searched using the following combined search terms: “PD-1 OR PD-L1” [Title/Abstract], “antibody” [Title/Abstract], “pharmacokinetic” and “monkey”. Only publications containing monkey PK data were included. In addition, references of relevant reviews were manually screened for further inclusion. For antibodies with monkey PK data available, human PK data were searched using the following combined search terms: “name of antibody” [Title/Abstract], “pharmacokinetic” [Title/Abstract], and “phase”. Antibodies without PK data reported in human were excluded. Only complete PK profiles after intravenous doses were included in this study to focus on the drug distribution and elimination phases.

The search was last updated on August 23, 2023. For the included antibodies, PK parameters were extracted from text or tables, and concentration-time curves were digitalized from figures using the WebplotDigitizer tool (<https://apps.automeris.io/wpd/>).

Prediction of Human PK Parameters Using Allometric Scaling

The allometry scaling is based on the hypothesis that a parameter can be scaled by body weight across species. The equation used is:

$$P_{human} = P_{animal} \times \left(\frac{BW_{human}}{BW_{animal}} \right)^b \quad (1)$$

where P represents a PK parameter (clearance, volume), BW is the body weight, and ‘b’ is the allometric exponent. The exponents used in this study were 0.85 and 1 for clearance (CL) and distribution volume (V), respectively.

Prediction of Human Concentration-Time Profiles by Species Time-Invariant Method

As for the STIV approach, the scaling is based on the assumption of time equivalency between two species and the assumption that the concentrations are correlated with body weight. The scaling is directly made on the concentration-time profiles using the following equations.

$$Time_{human} = Time_{animal} \times \left(\frac{BW_{human}}{BW_{animal}} \right)^{b_{volume} - b_{clearance}} \quad (2)$$

$$Concentration_{human} = Concentration_{animal} \times \frac{Dose_{human}}{Dose_{animal}} \quad (3)$$

where $Time_{human}$ is the equivalent time in human; $Time_{animal}$ is the observed time in animal; $Concentration_{human}$ is serum concentration of the drug in human; $Concentration_{animal}$ is serum concentration of the drug in animal; $Dose_{human}$ (mg/kg) is the dose in human PK study;

$Dose_{animal}$ (mg/kg) is the dose in animal PK study. The exponent ‘b’ is the allometric exponent, which is the same as the AS method.

Non-compartment PK parameters of the scaled concentration-time curves were calculated using R software (version 4.1.3, The R Foundation for Statistical Computing, <http://www.r-project.org/>) with “NonCompart” package.

Prediction of Human PK Through the PBPK Approach

The PBPK model was firstly developed in cynomolgus monkey to characterize the PK of antibodies in a relevant species. Modeling and simulation were performed by PK-Sim software (Open Systems Pharmacology Suite, <http://www.opensystemspharmacology.org>), version 11.0. The PBPK model for monoclonal antibodies represents an extension of the small molecule PBPK model as proposed by Niederalt et al.¹⁰ The generic model was used without further modifications. Briefly, the PBPK model in PK-Sim contains 15 organs/tissues, and incorporates the two-pore formalism for drug extravasation from blood plasma to interstitial space, lymph flow, drug catabolism in the endosomal space, as well as protection from catabolism by FcRn mediated recycling.¹⁰ In all models, the FcRn association rate (K_{ass}) was kept unchanged at the default value of $0.87 \text{ L} \cdot \mu\text{M}^{-1} \cdot \text{min}^{-1}$. The dissociation constant of drug for binding to FcRn (K_D) in endosomal space was optimized based on the observed data. No binding to FcRn was assumed in plasma and the interstitial space, and K_D in plasma/interstitial space was set to a very high value as default. This study focuses on the linear part of PK in order to compare the scaling results of the three methods with concentration data only. Therefore, target-mediated drug disposition (TMDD) was not taken into consideration in this study.

The model established in cynomolgus monkey was used to predict the PK profile in human. Firstly, the species in simulation was changed from “monkey” to “human”, and the physiological parameters changed accordingly. The body weights of Caucasian and Asian were assigned to be 70 kg and 60 kg, respectively, if they were not reported in the literature. The parameter K_D in endosomal space for human was assumed to be two-fold of that for monkey, since the affinity of cyFcRn was reported to be ~2-fold higher than that of hFcRn.^{15,16} Other parameters including K_D in plasma/interstitial space and K_{ass} in human PBPK model were consistent with those in monkey PBPK model.

Results

Data

A total of 260 papers were identified in the initial search of monkey PK data. After several circles of selection, six antibodies (avelumab, atezolizumab, nivolumab, pembrolizumab, cemiplimab, and zimberelimab) were included for investigation. Sasanlimab, CS1003, CDX-527, FS222, JS-001, and REGN3504 were excluded because those drugs only have monkey PK data available, but have no human PK data. Detailed descriptions on the characteristics of studies included in the monkey dataset are presented in Table 1. With regard to human concentration-time curves, only complete PK profiles after an intravenous single dose of the antibody were selected for this study to exclude potential sources of variability caused by accumulation. Papers with human PK data included in this study were summarized in Supplementary Table 1.

Prediction of Human PK Parameters Using Allometric Scaling

Human CL and V were scaled from the PK parameters of cynomolgus monkey. PK parameters of monkey were calculated using non-

Table 1

Published monkey PK of PD-1/PD-L1 antibodies included in this study.

Reference	Drug	Number of monkeys	Dose	CL (L/day/kg)	Vss (L/kg)	AUC _{0-t} (day·μg·mL ⁻¹)	AUC _{0-∞} (day·μg·mL ⁻¹)	Concentration-time curves
Jin et al. ¹⁷	Avelumab	N = 3	IV injection 5 mg/kg	0.0283	0.0976 ^a	175.36 ^a	184.05 ^a	Included
Jin et al. ¹⁷	Atezolizumab	N = 3	IV injection 5 mg/kg	0.0068 ^a	0.0419 ^a	687.89 ^a	739.32 ^a	Included
Deng et al. ¹⁸	Atezolizumab	N = 4 (2 females and 2 Males) per group	IV injection 0.5 mg/kg 5 mg/kg 20 mg/kg	0.0101 ^a 0.0045 ^a 0.0041 ^a	0.0769 ^a 0.0296 ^a 0.0595 ^a	29.0 ± 2.84 409 ± 56.3 1940 ± 192		Included
Wang et al. ¹⁹	nivolumab	N not known (male) N not known (female) N not known (male)	IV injection 1 mg/kg 1 mg/kg 10 mg/kg	0.0054 ± 0.0005 0.0060 ± 0.0009 0.0041 ± 0.0014	0.046 ± 0.005 0.052 ± 0.002 0.060 ± 0.016	167.08 ± 26.88 148.75 ± 23.88 1962.5 ± 516.7	186.28 ± 17.63 168.75 ± 25.67 2675 ± 1141.7	Not included
Zhang et al. ²⁰	nivolumab	N = 6 (3 males and 3 females) per group	IV injection 10 mg/kg	0.0038	0.0509	2396.56	2940.88 ^a	Included
Hutchins et al. ²¹	Pembrolizumab	N = 6 (females)	IV injection 0.3 mg/kg 3 mg/kg 30 mg/kg	0.0057 ± 0.0002 0.0042 ± 0.0004 0.0037 ± 0.0001	0.0309 ± 0.0060 0.0368 ± 0.0046 0.0548 ± 0.0057	41.0 ± 2.9 700.0 ± 76.5 6374 ± 767		Not included
Burova et al. ²²	cemiplimab	N = 5 (females) per group	IV injection 1 mg/kg 5 mg/kg 15 mg/kg	0.0060 0.0046 0.0037	0.0125 ^a 0.0121 ^a 0.0516 ^a	168 1100 3950	172.79 ^a 1186.26 ^a 4456.77 ^a	Included
Lou et al. ²³	Zimberelimab	N = 6 (3 males and 3 females) per group	IV injection 2 mg/kg 6 mg/kg 18 mg/kg	0.0069 ± 0.0009 0.0067 ± 0.0007 0.0044 ± 0.0007	0.0484 ± 0.0075 0.0494 ± 0.0065 0.0463 ± 0.0055	262.5 ± 55 887.5 ± 107.1 4087.5 ± 679.2	294.2 ± 42.5 908.3 ± 96.3 4208.3 ± 695.8	Included

The values are represented as mean or mean ± SD.

^a Calculated from concentration-time curves.**Table 2**

Observed and predicted human clearance and distribution volume for PD-1/PD-L1 antibodies using allometric scaling method.

Drug	CL _m (L/day/kg)	V _m (L/kg)	CL _{h,pred} (L/day/kg)	V _{h,pred} (L/kg)	CL _{h,obs} (L/day/kg)	V _{h,obs} (L/kg)
Atezolizumab	0.0064	0.0520	0.0041	0.0520	0.0029–0.0039	0.0512–0.0986
Avelumab	0.0283	0.0976	0.0184	0.0976	0.0076–0.0195	0.0499–0.0674
Pembrolizumab	0.0045	0.0408	0.0030	0.0408	0.0025–0.0036	0.0653–0.0857
Nivolumab	0.0048	0.0522	0.0031	0.0522	0.0030–0.0093	0.0596–0.1557
Cemiplimab	0.0048	0.0254	0.0031	0.0254	0.0030–0.0045	0.0569–0.0643
Zimberelimab	0.0060	0.0480	0.0039	0.0480	0.0034–0.0060	0.0510–0.0717

CL_m, observed mean value of monkey clearance; V_m, observed mean value of monkey distribution volume; CL_{h,pred}, predicted human clearance; V_{h,pred}, predicted human distribution volume; CL_{h,obs}, observed range of human clearance; V_{h,obs}, observed range of human distribution volume.

compartmental analysis (NCA) method if they were not reported in the paper. The body weight of cynomolgus monkey and human were assumed to be 4 kg and 70 kg, respectively. The results of AS are shown in Table 2. Human PK parameters were scaled from the reported monkey PK parameters, and the mean values for each drug were presented. Aside from the included paper, observed human PK parameters were also referred to the approved drug labels, resulting in the range of CL and V in Table 2. The predicted CLs were within the observed range, except for atezolizumab. However, the majority of Vs were not well predicted by AS. The distribution volume of avelumab was overpredicted, while V of pembrolizumab, nivolumab, cemiplimab, and zimberelimab were underpredicted.

Prediction of Human Concentration–Time Profiles by Time-Invariant Method

Human serum concentration-time curves predicted by the time-invariant method were shown in Fig. 1. Body weight of cynomolgus monkey was assumed to be 4 kg. Body weight of human was assigned to 70 kg and 60 kg for Caucasian and Asian, respectively, if not reported. PK profiles of cemiplimab in human could be well scaled from those in monkey (Fig. 1 A). Concentration-time curves of zimberelimab in human predicted

using monkey data of a low (2 mg/kg) or middle (6 mg/kg) dose were consistent with the observed profiles, but those scaled from monkey data of a high dose (18 mg/kg) were overpredicted after 200 h. For nivolumab and avelumab, after administration of a single dose of 1 mg/kg antibody, PK curves in human could be well scaled from monkey data, while PK profiles for a higher dose could not be well predicted. It should be noted that nivolumab in Caucasian was especially overpredicted in the elimination phase (Fig. 1 D). Pharmacokinetics of atezolizumab in human could be extrapolated from monkey PK in general. Concentration-time curves after administration of 5 mg/kg atezolizumab in monkey showed better predictive performance than the curves after administration of a higher or a lower dose (Fig. 1 F and G).

Prediction of Human PK Through PBPK Approach

PBPK models of the five antibodies were established to fit the concentration-time curves in cynomolgus monkey firstly. The key parameter K_D in endosomal space describing the affinity of antibody binding to cynomolgus FcRn was fitted to the observed data, and the value for atezolizumab, avelumab, nivolumab, cemiplimab, and zimberelimab were 0.4 μmol/L, 3.26 μmol/L, 0.19

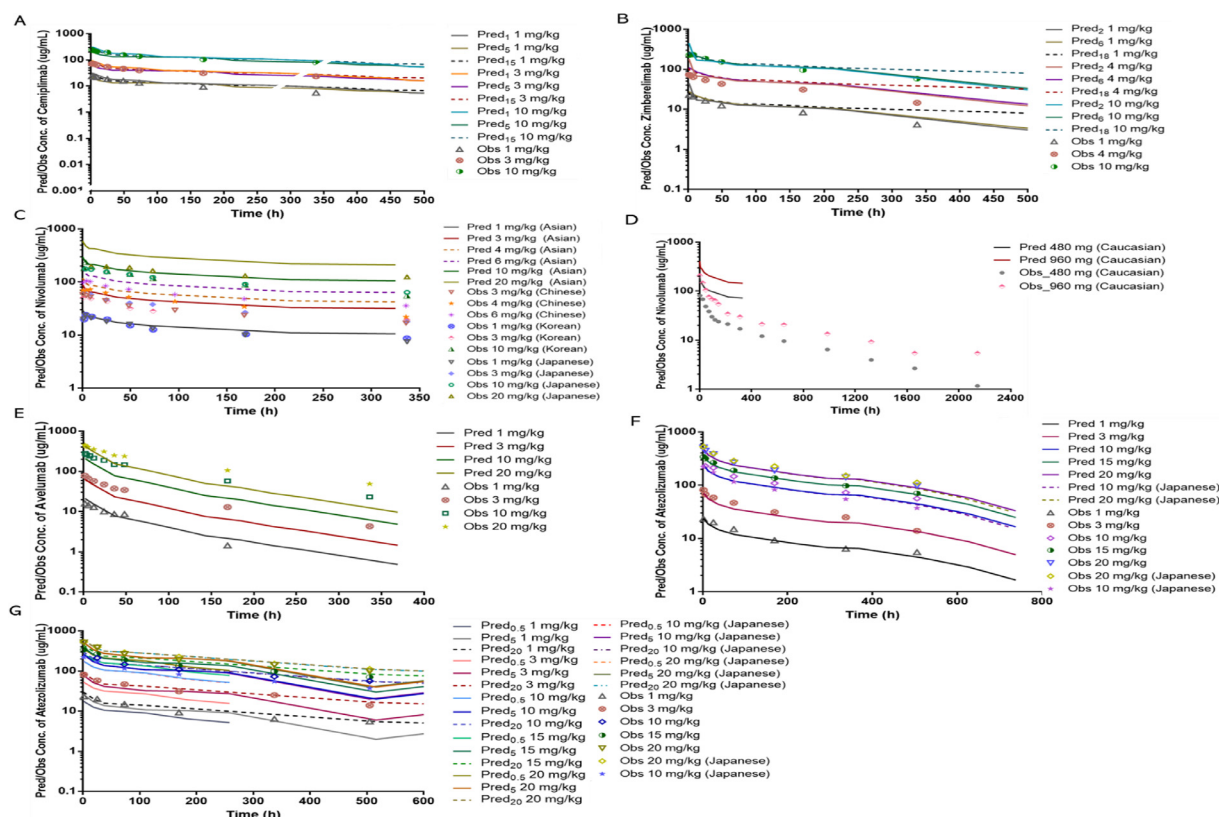


Figure 1. Human serum concentration-time curves predicted by the species time-invariant method. Lines represent the predicted curves. Symbols are the observed data. Pred; j mg/kg, i means the dose of antibodies used for scaling in monkey, j is the dose of antibodies in human. A. Cemiplimab, B. Zimberelimab, C. Nivolumab predicted in Asian, D. Nivolumab predicted in Caucasian, E. Avelumab, F and G. Atezolizumab.

$\mu\text{mol/L}$, $0.33 \mu\text{mol/L}$, and $0.65 \mu\text{mol/L}$, respectively. The predicted and observed curves in monkey are shown in [Supplementary Fig. 1](#). The model could capture the linear part of PK profiles of antibodies in monkey, without regard to TMDD and anti-drug antibodies (ADA). The PBPK model developed in monkey was used to predict PK in human. Expect that K_D in endosomal space for human was two-fold of that for monkey, other drug-specific parameters were the same in human and monkey models. Human concentration-time curves predicted by PBPK models were presented in [Fig. 2](#). The model showed good prediction of the PK of atezolizumab, cemiplimab, and zimberelimab. For avelumab, overprediction was observed at a low (1 mg/kg) dose. The overprediction may due to TMDD which was not integrated in the model. However, underpredictions of concentration were shown at higher doses, indicating that K_D of $6.52 \mu\text{mol/L}$ for avelumab in human was overpredicted. On the contrary, K_D for nivolumab was underpredicted in human, especially for Caucasian, as shown in [Fig. 2 F](#).

Discussion

Translational PK of therapeutic proteins is important for drug development. In this study, we compared the three methods in predicting the PK of PD-1/PD-L1 antibodies in human. A generic PBPK model in PK-Sim was used, and no further mechanism such as TMDD was included in order to match the quantity of data and time taking for the three methods. The three methods used in this study were all simple, quick, and resource-saving.

Because FcRn is highly conserved between monkeys and humans, AS from monkeys to humans is the simplest and most robust approach for predicting CL of antibodies at the linear PK region. It has

been demonstrated that single-species AS using cynomolgus monkeys showed better prediction accuracy of human PK than multiple-species scaling.^{14,24} Different exponents of AS from monkey to human have been previously reported for therapeutic proteins. The scaling exponent for CL was reported to be in the range of 0.7–1.0, whereas the value of the exponent changed slightly depending on the dataset used for analyses.^{25,26} In this study, we investigated the exponent of 0.75 and 0.85 for CL. A better correlation was obtained between the observed human CL and the estimated human CL from cynomolgus monkey PK data based on an exponent of 0.85 for CL compared with an exponent of 0.75. Therefore, 0.85 was chosen as the AS exponent for CL finally. Accordingly, human concentration-time profiles were also predicted from the cynomolgus monkey data using STIV method with a fixed exponent of 0.85 for CL and 1.0 for V in our study.

It has been reported that the STIV method can provide higher accuracy of prediction than simple dose-normalization, regardless of the type of target antigens (soluble or membrane-bound).²⁷ Aside from linear PK, it has also been found that the STIV method and minimal PBPK model can be employed to reliably translate non-linear PK of monoclonal antibodies from cynomolgus monkey to human.^{8,13} In our study, STIV and generic PBPK method showed comparable prediction accuracy. We found that the prediction of STIV using cynomolgus monkey PK data of a middle dose was better than that of a low or a high dose. We calculated the fold errors (predicted value/observed value) of CL and V derived from STIV and PBPK approach for each extracted PK curve in human, and the results were shown in [Fig. 3](#). Compared with the generic PBPK model, human CLs were better predicted by the STIV method. However, the generic PBPK model showed better performance on the prediction of distribution volume compared

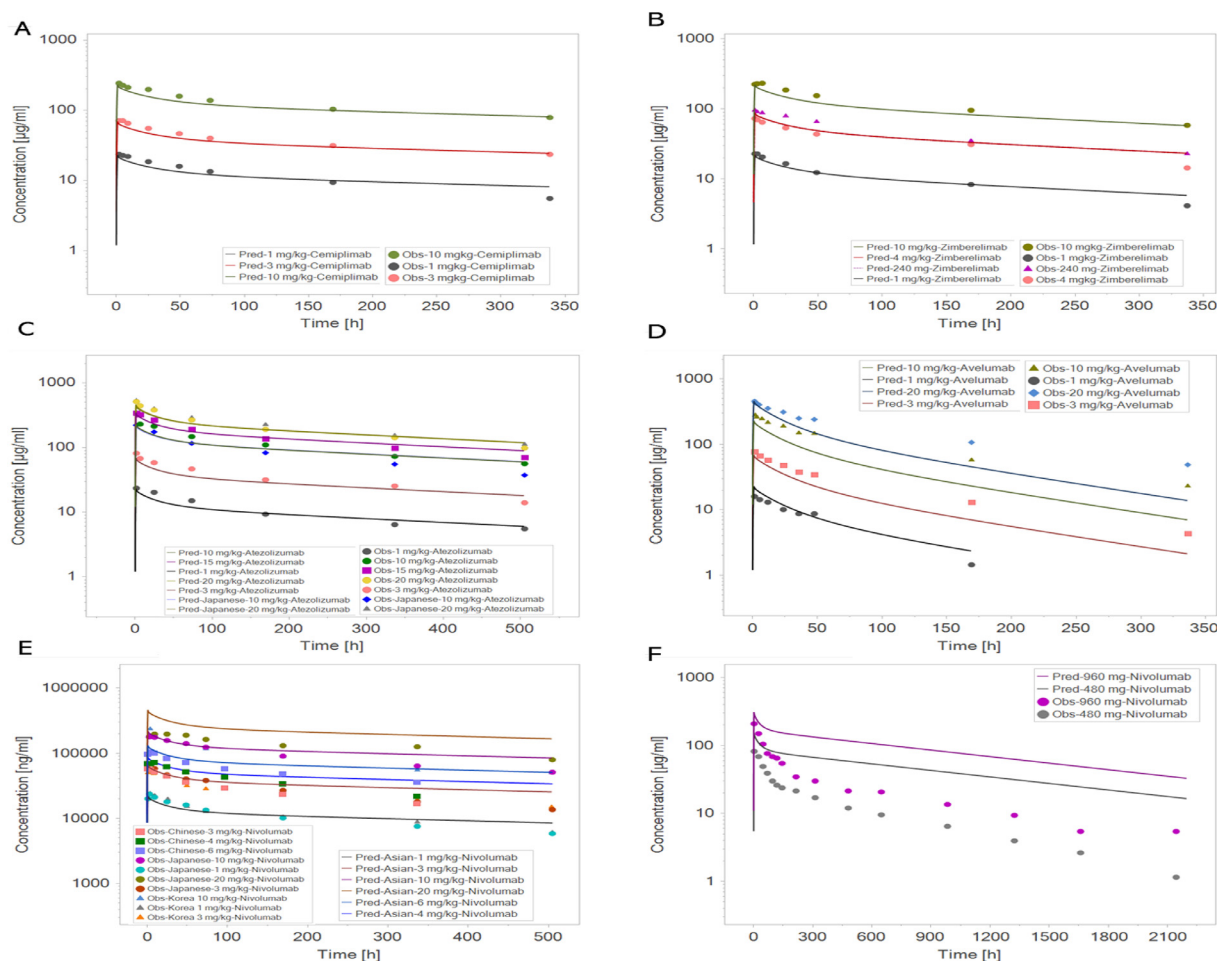


Figure 2. Human concentration-time curves predicted by PBPK model. Lines represent the predicted curves (Pred). Symbols are the observed data (Obs). The simulated population is Caucasian if not specified. A. Cemiplimab, B. Zimberelimab, C. Atezolizumab, D. Avelumab, E. Nivolumab predicted in Asian, F. Nivolumab predicted in Caucasian.

with the STIV method. It should be noted that STIV method depends heavily on the observed data points. The time predicted for human was extrapolated from the monkey data. As shown in Fig. 1 D, PK profiles in human after 400 h could not be predicted because of the limited monkey data. This is one of the limitations of STIV method. In contrast, the PBPK model does not have this limitation.

Although a generic PBPK model frame was used, the prediction of CL was rather empirical since K_D in endosomal space for human was set to be two-fold of that for monkey. The same K_D in endosomal space for human and cynomolgus monkey was also tested in this study. Avelumab was better predicted when the same K_D in endosomal space was used for human and monkey. The CL of avelumab was higher compared with other antibodies, the two-fold scaling of K_D in endosomal space may be not applicable for antibodies with high CL such as avelumab. The predictions may be more accurate if the experimental data of K_D were available. The generic PBPK model in PK-Sim was employed in this study. In general, for most marketed antibodies, in order to achieve optimal efficacy, the drug exposure is usually much larger than the amount of the target. In this case, the impact of target-mediated clearance on total antibody clearance is relatively low, and the non-specific clearance is basically linear within the therapeutic dose range. It does not mean that mechanism components such as TMDD are not necessary. With more mechanistic

incorporation involved, the PBPK model could potentially make more precise prediction and work for more complex situation. The significance of the mechanistic PBPK model should not be ignored. However, more data and time are needed to develop such a mechanistic model (Fig. 3).

There are some limitations in this study. Firstly, the predictions heavily depend on the PK data of cynomolgus monkey. Since the data of monkey were extracted from literature, bias could not be avoided. In some cases, different PK profiles were reported by different paper, even if the same dose was given. While for nivolumab, only one PK curve in monkey was available, which makes the prediction by time-invariant and PBPK method less reliable. Furthermore, the antibodies were randomly selected based on the availability of cynomolgus monkey and human PK data in this study, only five antibodies were investigated for all the three methods. Further studies are still needed.

In summary, translating human PK of PD-1/PD-L1 antibodies from cynomolgus monkey using empirical AS method, the STIV approach, and the generic PBPK model in PK-Sim were investigated. Allometric scaling could be used to predict human CL, but was not expected to predict distribution volume. The STIV method and generic PBPK model can be employed to translate PK profiles of PD-1/PD-L1 antibodies from cynomolgus monkey to human with comparable efficacy.

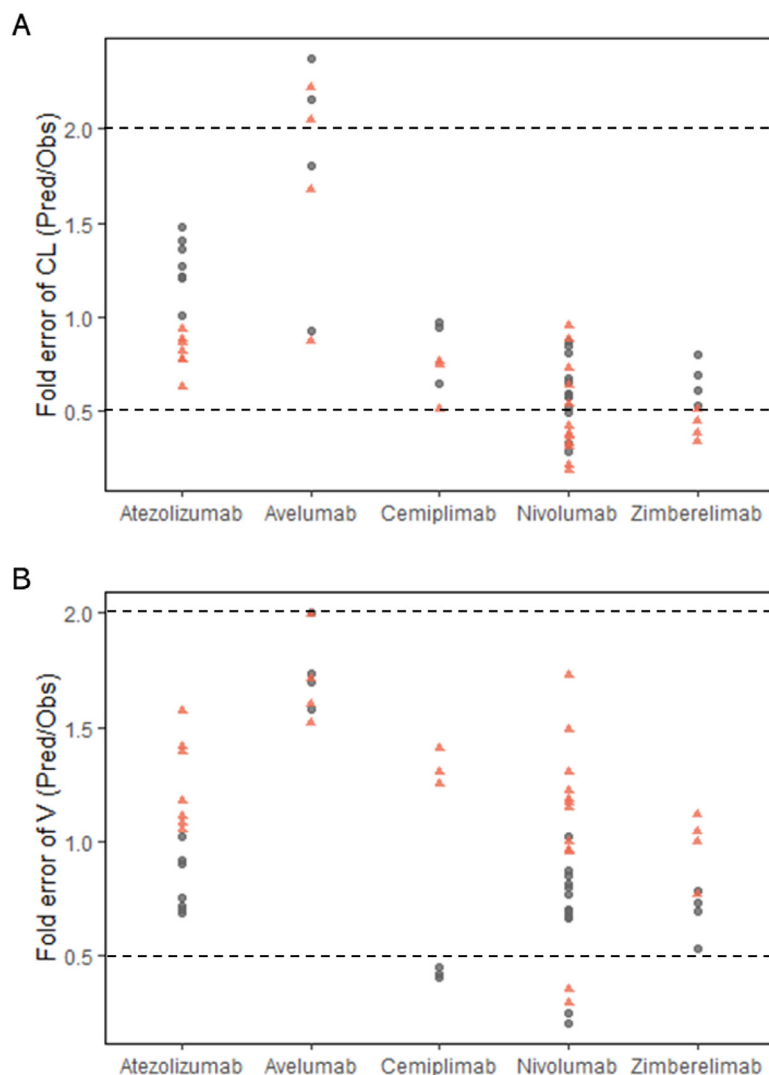


Figure 3. Fold errors of predicted/observed clearance (CL) and distribution volume (V) for antibodies in human using STIV and PBPK methods. Grey dots are the fold errors of STIV method, and red triangle represent fold errors using PBPK method.

Data Availability

The data of this study are publicly available.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.xphs.2024.07.003](https://doi.org/10.1016/j.xphs.2024.07.003).

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Supplementary Materials

Supplementary Table 1 Published human PK of PD-1/PD-L1 antibodies included in this study.

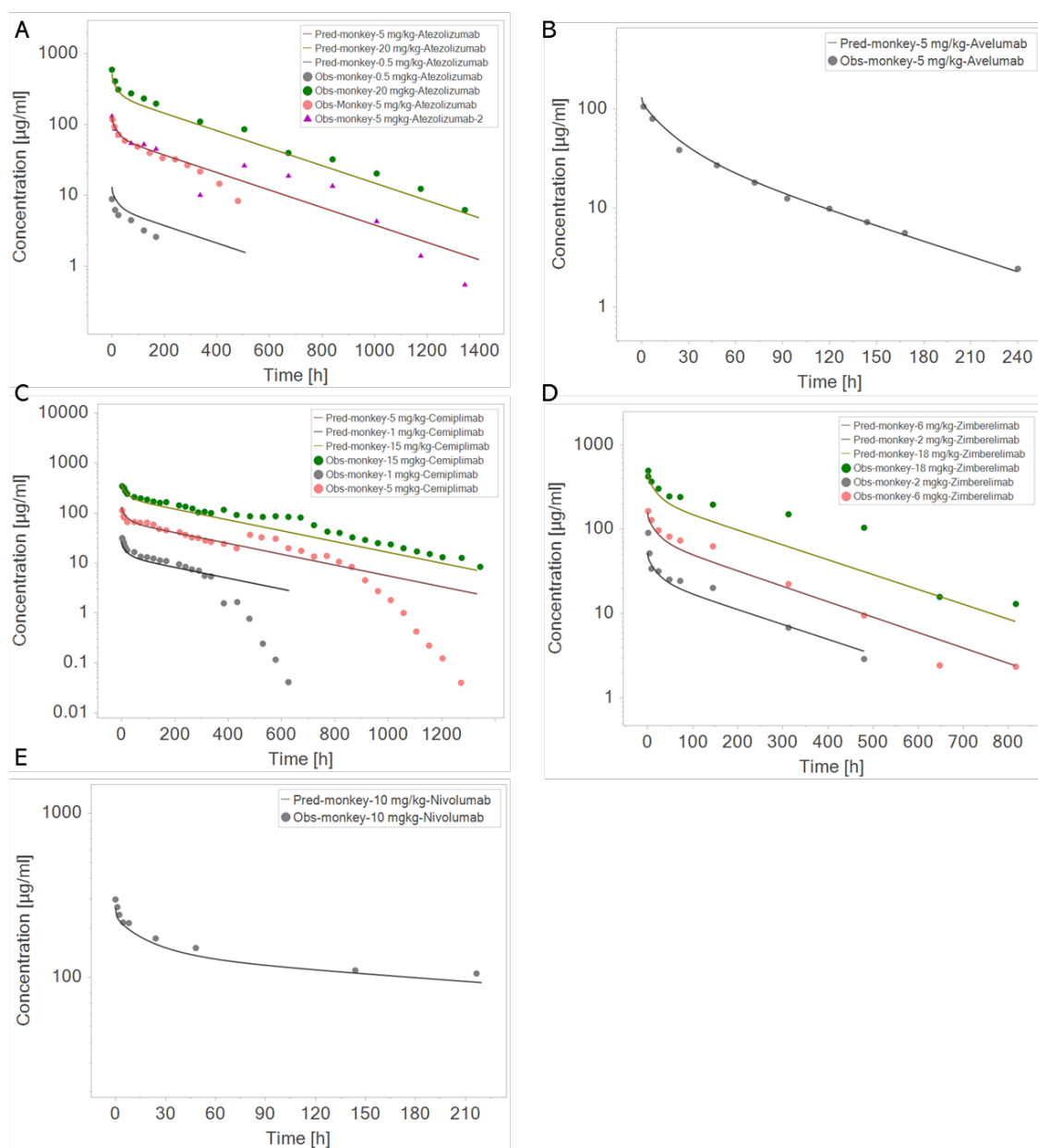
Reference	Drug	Number of subjects	Dose	CL (L/day/kg)	Vss (L/kg)	AUC0-t (day·µg·mL ⁻¹)	AUC0-∞ (day·µg·mL ⁻¹)	Concentration-time curves
Lee et al. ¹	Nivolumab	6	IV infusion					Included
		6	(over 60 minutes)	0.0030	± 0.0664	± 212.92	± 382.5	
		6		0.0012	0.0106	39.67	130.83	
		Korean	1 mg/kg	0.0036	± 0.0860	± 508.33	± 950 ± 416.67	
			3 mg/kg	0.0012	0.0185	85.83	2837.5	
			10 mg/kg	0.0051	± 0.0911	± 1612.5	± 1445.83	
				0.0043	0.0555	591.67		
Hotchkiss et al. ²	Nivolumab	11	IV infusion	0.0086	0.1557	754.13	790.04	Included
		11	480 mg	0.0093	0.1485	1338.75	1507.92	
		USA	960 mg					
Yamamoto et al. ³	Nivolumab	3	IV infusion					Included
		5	1 mg/kg	0.0030	± 0.0646	± 206.25	± 333.33	
		6	3 mg/kg	0.0005	0.0067	24.17	57.92	
		3	10 mg/kg	0.0050	± 0.0697	± 512.5 ± 187.5	833.33	
		Japanese	20 mg/kg	0.0036	0.0102	1829.17 ± 300	470.83	
				0.0030	± 0.0836	± 2808.33	± 3445.83	
				0.0006	0.0274	645.83	470.83	
				0.0049	± 0.0968	±	5250±2583.33	

				0.0034	0.0121			
Ma et al. ⁴	Nivolumab	15	3 mg/kg	0.0044 ^a	0.0803 ^a	377.51 ^a	686.07 ^a	Included
		20	240 mg	0.0040 ^a	0.0596 ^a	533.69 ^a	862.19 ^a	
		11	360 mg	0.0034 ^a	0.0719 ^a	757.89 ^a	1507.72 ^a	
Shimizu et al. ⁵	pembrolizumab	Chinese						Included
		3	IV infusion					
		6-7	(over 30 minutes)	0.0025	0.0653	507	813	
Heery et al. ⁶	Avelumab	Japanese	2 mg/kg	0.0029	0.0765	2219	3410	Included
			10 mg/kg					
		4	IV infusion			43.33 ± 18.45	53.75 ± 27.08	
		13	(over 60 minutes)			253.33 ±	285.42 ± 87.5	
		48		0.0195 ^a	0.0587 ^a	82.08	1191.67	
		21	1 mg/kg	0.0100 ^a	0.0575 ^a	1033.33 ±	± 320.83	
		USA	3 mg/kg	0.0076 ^a	0.0499 ^a	381.25	1941.67	
Herbst et al. ⁷	Atezolizumab		10 mg/kg	0.0084 ^a	0.0633 ^a	1879.17 ± 650	± 770.83	Included
			20 mg/kg					
		USA	IV infusion					
			1 mg/kg	0.0029 ^a	0.0781 ^a	202.63 ^a	353.18 ^a	
			3 mg/kg	0.0034 ^a	0.0512 ^a	663.17 ^a	871.15 ^a	
			10 mg/kg	0.0031 ^a	0.0592 ^a	2170.65 ^a	3235.73 ^a	
			15 mg/kg	0.0034 ^a	0.0734 ^a	2827.49 ^a	4315.21 ^a	
Mizugaki et al. ⁸	Atezolizumab		20 mg/kg	0.0033 ^a	0.0692 ^a	3999.02 ^a	6071.64 ^a	
		3	IV infusion					
		3	10 mg/kg	0.0039	± 0.0620	±	2290 ± 101	
		Japanese	20 mg/kg	0.0010	0.0190		6630 ± 668	

				0.0036	±	0.0637	±		
				0.0010		0.0120			
Papadopoulos et al. ⁹	Cemiplimab	18	IV infusion (30 minutes)						Included
		36							
		6	1 mg/kg	0.0045 ^a		0.0569 ^a		151.85 ^a	221.91 ^a
	USA		3 mg/kg	0.0030 ^a		0.0643 ^a		494.72 ^a	997.53 ^a
			10 mg/kg	0.0031 ^a		0.0623 ^a		1668.98 ^a	3244.78 ^a
Liu et al. ¹⁰	Zimberelimab		IV infusion						Included
		3	1 mg/kg	0.006 ± 0.002		0.0717	±	126.88	± 183.58
		3	4 mg/kg	0.006	±	0.0283		61.79	95.54
		4	10 mg/kg	0.0012		0.0717	±	442.58	± 620.63
		8	240 mg Q2W	0.0052	±	0.0233		130.38	140.25
		6	240 mg Q3W	0.0016		0.065 ± 0.02		1347.17	± 2339.08±
	Chinese			0.0052	±	0.0517	±	461.46	907.92
				0.0024		0.0133		597.25	± 876.79
				0.0044	±	0.0567 ± 0.02		232.13	328.42
				0.0016				767.5	± 946.92
								195.96	240.96

The values are represented as mean or mean ± SD.

^a Calculated from concentration-time curves.



Supplementary Figure 1 Observed and PBPK model predicted concentration-time curves of cynomolgus monkey. A. Atezolizumab, B. Avelumab, C. Cemiplimab, D. Zimberelimab, E. Nivolumab.

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